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رقم الكمبيوتر *

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الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) *

WKBB

 Will be reviewed

8

If a process must loop repeatedly, keeping the CPU occupied, to wait for a condition, that is called: 

☐ starvation☐ livelock

- ☐ deadlock
- ☒ busy waiting
- ☐ mutual exclusion
- ☐ coroutine

✓ **Correct** 1/1 Points

9

Given the following processes and burst times. The processes are assumed to have arrived all at time 0.

What is the turnaround time of process P5 using the Round Robin (RR, quantum = 2 milliseconds)?



17 milliseconds

✓ **Correct** 1/1 Points

10

Given the following processes and burst times. The processes are assumed to have arrived all at time 0.

What is the waiting time of process P5 using the Round Robin (RR, quantum = 2 milliseconds)?



12 milliseconds

✉ **Will be reviewed**

11

In Readers & Writers problem, can the writers suffer from starvation? 

☒ Yes

☐ No

✉ **Will be reviewed**

12

Preemptive scheduling allows a process to be interrupted in the midst of its execution, taking the CPU away and allocating it to another process 

☒ True

☐ False

✓ **Correct** 1/1 Points

13

Consider the following snapshot of a system:

What is the content of the matrix Need?



Need = Max - Allocation

	A	B	C	D
P0	0	0	0	0
P1	0	7	5	0
P2	1	0	0	2
P3	0	0	2	0
P4	0	6	4	2

✗ **Incorrect** 0/1 Points

14

Consider the following snapshot of a system:

If a request from process P1 arrives for (0, 4, 2, 0), can the request be granted? If the request is granted, write a safe sequence.



Can't accept the request because system will be in an unsafe state & P1 must wait.

✉ **Will be reviewed**

15

Given the following processes and burst times. The processes are assumed to have arrived all at time 0.

Which of the algorithms results in the minimal average waiting time.

☒ SJF

☐ RR

 **Will be reviewed**

16

'Aging' is 

- ☐ keeping track of cache contents
- ☐ keeping track of what pages are currently residing in the memory
- ☐ keeping track of how many times a given page is referenced
- ☒ increasing the priority of jobs to ensure termination in a finite time.

✓ **Correct** 1/1 Points

17

Given the following processes and burst times. The processes are assumed to have arrived all at time 0.

What is the turnaround time of process P5 using the Shortest Job First (SJF)?



11 milliseconds

✓ **Correct** 1/1 Points

18

Below the structure of the Readers & Writers problem is shown using the semaphores mutex & wrt and an integer variable readCount.

What are the initial values of mutex, wrt, and readCount?



mutex = 1
wrt = 1
readCount = 0

Will be reviewed

19

Suppose that a process is in "BLOCKED" state waiting for some I/O service. When the device is completed, it goes to the

- ☐ BLOCKED state
- ☒ READY state
- ☐ SUSPENDED state
- ☐ TERMINATED state

Will be reviewed

20

In the readers/writers problem, we want to allow a reader and a writer to be in their critical sections for the shared data at the same time.

- ☐ True
- ☒ False

✉ Will be reviewed

21

If process A is waiting for a nonpreemptible resource that is held by process B and process B is waiting for a nonpreemptible resource that is held by A, that is called: 

- ☐ starvation
- ☐ livelock
- ☒ deadlock
- ☐ busy waiting
- ☐ mutual exclusion
- ☐ coroutine

✓ **Correct** 1/1 Points

22

Consider the following snapshot of a system:

Is the system in a safe state? If the system is safe, write a safe sequence.



< P0 , P2 , P1 , P3 , P4 >

✉ Will be reviewed

23

An unsafe state may not necessarily lead to deadlock 

☒ True☐ False **Will be reviewed**

24

In the producer/consumer problem the producer never has to wait. 

☐ True☒ False **Will be reviewed**

25

If processes A, B, and C are all requesting resource R, but the operating system allows A and B to access R repeatedly while C continues to wait for R, that is called: 

☒ starvation☐ livelock☐ deadlock☐ busy waiting☐ mutual exclusion

☐ coroutine

 **Will be reviewed**

26

A situation where several processes access and manipulate the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place, is called a 

- ☐ convoy effect
- ☐ disorder problem
- ☐ critical-section problem
- ☒ race condition

 **Will be reviewed**

27

A binary semaphore is a semaphore with integer values 

 ☒ 1

- ☐ -1
- ☐ 2
- ☐ -2